



Assessing the information security posture of online public services worldwide: Technical insights, trends, and policy implications[☆]

Diogo Ribeiro^{a,b,*}, Vítor Fonte^{b,c}, Luis Felipe Ramos^b, João Marco Silva^{a,b}

^a INESC TEC, Rua Dr. Roberto Frias, 4200-465 Porto, Portugal

^b University of Minho, Rua da Universidade, 4710-057 Braga, Portugal

^c UNU-EGOV, Rua de Vila Flor 166, 4810-225 Guimarães, Portugal

ARTICLE INFO

Keywords:

EGOV
Information security
Cybersecurity
Public services

ABSTRACT

The fast global expansion of online public services has transformed how governments interact with citizens, offering convenience and efficiency. However, this digital transformation also introduces significant security risks, as sensitive data exchanged between users and service providers over public networks are exposed to cyber threats. Thus, ensuring the security and trustworthiness of these services is critical to the success of Electronic Government (EGOV) initiatives. This study evaluates the information security posture of 3068 public service platforms across all 193 UN Member States through non-intrusive assessments conducted in 2023 and 2024. The evaluation focuses on three key dimensions: (i) the adoption of secure end-to-end communication protocols, (ii) the trustworthiness of digital certificate chains, and (iii) the exposure of hosting servers to known vulnerabilities. The findings reveal that while some progress has been made in securing online public services, substantial gaps remain in the implementation of international security standards and best practices. Many platforms continue to rely on outdated cryptographic protocols, misconfigured certificates, and unpatched vulnerabilities, leaving citizens and services vulnerable to cyber threats due to weaknesses that malicious actors can easily and inconspicuously identify. These insights emphasize the need for effective implementation of more comprehensive cybersecurity policies, proactive security assessments, and improved regulatory compliance checks. Additionally, this work provides actionable guidance for governments and system administrators to enhance the security of EGOV infrastructures by addressing persistent vulnerabilities and adopting robust cybersecurity practices.

1. Introduction

Throughout the last few decades, there has been a consistent effort by public administrations worldwide towards enhancing the coverage and efficiency of services provided to citizens, mainly through investment in Information and Communication Technologies (ICT). The modernisation of Electronic Government (EGOV) promotes government transparency and accountability whilst mitigating information asymmetries in the society (Alarabiat et al., 2018; Coleman, 2008).

The United Nations Department of Economic and Social Affairs (UN DESA) performs a biennial evaluation on the development status of EGOV across all UN Member states. The UN E-Government Survey measures countries' performance and their progress towards achieving the Sustainable Development Goals (SDGs), identifying existing challenges and areas of strength, ultimately aiding in shaping their policies

and strategies (UN DESA, 2022).

A significant oversight in the UN DESA survey is the lack of an assessment concerning both the current state and the evolving trends of information security and privacy within platforms supporting these services. The communication between EGOV service providers and citizens occurs through public communication infrastructures exposed to a multitude of threats. Given the sensitive nature of data being exchanged and the spike in malicious activities undermining digital trust (Chen et al., 2023), it is imperative to adopt the best practices and technologies on the server side in a way that ensures a secure and trustworthy delivery of online public services.

Hence, this work provides a worldwide overview of the information security posture of EGOV services connecting public administration to citizens. The security posture is assessed by evaluating three key aspects of public-facing online services: (i) the support of up-to-date and robust

[☆] This article is part of a Special issue entitled: 'HICSS2024 SI' published in Government Information Quarterly.

* Corresponding author at: INESC TEC, Rua Dr. Roberto Frias, 4200-465 Porto, Portugal.

E-mail addresses: diogo.p.ribeiro@inesctec.pt (D. Ribeiro), vtor.fonte@unu.edu (V. Fonte), joaomarco@di.uminho.pt (J.M. Silva).

protocols for secure end-to-end communications; (ii) the trustworthiness of the digital certificate chain of the provided services; and (iii) the exposure of the hosting servers to known vulnerabilities.

The analysis highlights the risks of compromising the security properties of confidentiality, integrity, availability, and authenticity due to flawed service configurations or inadequate patching and security policies. Although this work does not extensively explore the underlying causes of the observed trends nor the sensitivity of the services hosted on the analyzed domains, it provides valuable insights for system administrators towards hardening infrastructures currently supporting EGOV solutions. Moreover, this study might have significant implications for cybersecurity policy-making and enforcement. By identifying gaps and discussing best practices, the findings offer actionable guidance for governments to develop more effective regulatory frameworks, minimize risk exposure, and improve resilience against cyber threats.

This study extends the previous work (Silva et al., 2024) by considering the security trends of 3068 platforms from all UN Member States over one year (i.e., between 2023 and 2024). These platforms serve as strategic hubs for government ministries, federal agencies, and national portals, facilitating public service delivery.

The remaining sections in this article are organized as follows: [Section 2](#) presents previous work in the field of information security in EGOV services and their relation with cybersecurity policymaking. [Section 3](#) introduces the methodology used in this work to perform the evaluation of the security posture of EGOV service providers, while presenting the main technologies that support the trust model of end-to-end secure communications. [Section 4](#) discusses the various research outcomes and trends. [Section 5](#) discusses the key findings and suggests actions to improve information security in the surveyed platforms, while [Section 6](#) concludes the article highlighting its practical implications, limitations, and pointing directions to future research on the information security posture of EGOV platforms.

2. Related work

Beyond effectiveness, coverage, and usability, the successful adoption of EGOV solutions is closely related to the level of trust citizens can put in the provided services. This trustworthiness can be promoted through the perceived levels of security and privacy that services and underlying infrastructure can ensure. However, since implementation details are often not publicly disclosed, assessing whether EGOV service providers follow best practices and adopt appropriate technologies remains challenging. Additionally, most citizens lack the expertise to evaluate these aspects themselves. In the end, the answer to these challenges typically falls into the government's capacity to design and enforce effective cybersecurity policies.

Over the last two decades, scientific literature has delved into the expansion of online public services and how new technologies, such as cloud computing (Susanto & Almunawar, 2016), artificial intelligence (Horowitz et al., 2018), and blockchain (Alketbi et al., 2018), and the underlying cyber threats are shaping global policy-making. Most studies adopt a theoretical approach towards secure online services (Li et al., 2024; Shayganmehr et al., 2023; Tegethoff et al., 2023) or rely on interviews and questionnaires to understand how public entities face cybersecurity-related issues (AlGhamdi et al., 2022; Ali et al., 2020). Although valuable, these approaches might not accurately reflect how systems are deployed and managed.

A more practical approach has been the adoption of evidence-based policy-making (Pawson, 2002), in which incident reporting (Busetti & Scanni, 2025; de Bruijn & Janssen, 2017) and cyber threat intelligence (Ainslie et al., 2023; Conti et al., 2018) are presented as strategic tools for policy shaping. Although several legislations, including the EU NIS2 Directive, the USA Cybersecurity Information Sharing Act, and the UK Data Protection Act, have established cybersecurity information sharing, its full potential is hindered by the reactive nature of incident reporting and the reluctance of many countries to fully embrace it.

Coupled with capacity gaps and the challenge of extracting meaningful insights from the ever-growing threat intelligence reports, effectively translating lessons into policy remains difficult (Busetti & Scanni, 2025).

A promising complementary approach is to rely on systematic cybersecurity assessments of infrastructures providing EGOV services as a proactive source of evidence for policymakers and as actual indicators of the effectiveness of implemented strategies. Since these assessments primarily evaluate how digital services are exposed to the Internet, they can overcome the challenges associated with non-public implementation details.

While studies have explored various aspects of information security assessments of EGOV services and infrastructure deployments, they typically focus on a single country or region. As a result, they fail to offer a comprehensive view of the global security posture adopted by the public sector, grounding the need for a broader assessment of current trends in this topic. For instance, Alsmadi and Shanab (2016) and Ali and Zamri Murah (2018) conducted penetration tests on platforms hosted on selected domain names in Jordan and Libya, respectively. The work presented by Thompson et al. (2020) audits a selection of 40 websites from Australia and Thailand to identify existing security vulnerabilities. Csontos and Heckl (2021) performed an analysis on the security posture of public services in Hungary, whereas Silva & Fonte (2019) analyzed the adoption of Secure Sockets Layer (SSL) and Transport Layer Security (TLS) protocols adoption in Portugal.

This study advances the literature on EGOV cybersecurity by comprehensively analysing current trends among providers worldwide and assessing their posture regarding the adoption of established technologies and guidelines. Additionally, it evaluates the evolving exposure to cyber threats, providing key insights and recommendations to strengthen service security and protect citizens. These results highlight persistent vulnerabilities and common misconfigurations, offering policymakers guidance on specific areas where regulatory frameworks should be more comprehensive, where proactiveness increases their effectiveness, and where it enhances cyber threat intelligence.

3. Methodology

This work provides a worldwide overview of how the public sector addresses security properties such as confidentiality, integrity, and trustworthiness in public-facing online services. By employing non-intrusive scanning methods, it evaluates providers' security posture in communication between offered services and citizens. This analysis is particularly relevant given that such interactions typically occur over public networks, which are exposed to numerous cyber threats.

The study analyzed 3068 unique domains from all 193 UN Member States hosting online platforms from government ministries, federal agencies, public service platforms, and national portals. The public dataset of domain names was provided to the United Nations E-Government Knowledge Database by national officials within the scope of the UN E-Government Survey (UN DESA, 2022).

To capture the evolution and trends in the security posture of platforms hosting worldwide strategic services, a longitudinal approach was adopted. An initial assessment was conducted from December 2022 to January 2023, followed by a second assessment from February to March 2024. This timeframe covers the year following the release of the 2022 UN E-Government Survey (UN DESA, 2022) and the period immediately after data collection for the 2024 edition (UN DESA, 2024). The aim is to identify whether efforts to improve global EGOV index rankings have motivated countries to implement security enhancements.

3.1. Platforms analyzed

In order to provide a geo-economic perspective to the assessment, the results in [Section 4](#) are presented taking into consideration domains' region and countries' income according to the World Bank's economy classification. This is the same approach adopted by the UN E-

Government Surveys, allowing a direct comparison between EGOV indexes and security posture (this comparative analysis is outside this work scope). Fig. 1 presents the geo-economic distribution of the analyzed domains.

Table 1 provides a more detailed view of the online platforms across the 22 sub-regions used in the UN Geoscheme. Due to privacy and security concerns, this work does not reference specific domains or country names, instead presenting the outcomes of the analysis as aggregates. This table also shows that the number of reachable domains decreased by 123 from the first to the second assessment. It means that these domains were offline during the evaluation period despite multiple scanning attempts (at least five on different days). As this study examines the observed trends in public services accessible to citizens, the discussion in Section 4 focuses exclusively on the domains that were accessible in both assessments.

3.2. Security analysis

Similarly to most online services, data exchange between citizens and public service providers typically takes place over public networks. In these networks, information security cannot be assured, and consequently, it is exposed to a wide range of threats. Given this constraint, it is critical to ensure two key requirements: (i) guaranteeing the authenticity of the entity providing a service, and (ii) protecting the confidentiality and integrity of the data being transmitted. These requirements are typically addressed through cryptography-based solutions (e.g., digital certificates and message authentication), alongside ensuring that hosting servers are not exposed to software vulnerabilities.

Thus, the assessments conducted in this work consist of inspecting the security posture of the online platforms hosted across all domains in the following three dimensions.

3.2.1. Support for secure end-to-end communications

The TLS protocol, currently in version 1.3 (Rescorla, 2018), is designed to establish secure communication between a client (e.g., a user's device) and a server (e.g., an online service). During the setup of this secure connection (i.e., *TLS handshake*), the client and server exchange information about the cryptographic algorithms they support. Based on this, they agree on which algorithms to use and generate the necessary encryption keys. By relying on asymmetric cryptography and key derivation methods, as long as the selected cryptographic algorithms are strong, TLS ensures that data shared between the client and server remains confidential and protected from tampering, even if the network itself is insecure or unreliable.

Regarding this dimension, an analysis on the adoption of newer and more secure cryptographic protocols is presented. Earlier versions of these protocols, such as SSLv2, SSLv3, TLSv1.0, and TLSv1.1, have well-documented security vulnerabilities that attackers can exploit. As a

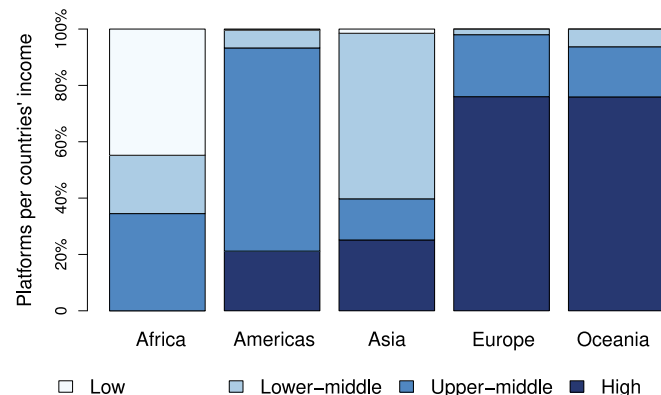


Fig. 1. Platforms per country region and income.

result, the Internet Engineering Task Force (IETF) has officially classified these versions as deprecated (Housley, 2019). Furthermore, it is also analyzed whether servers use *weak* or *insecure* encryption algorithms (i.e., ciphers). *Weak* ciphers refer to algorithms that, while not yet compromised, have design flaws, unexploited vulnerabilities, or use key lengths too short to provide adequate protection against modern attacks. In contrast, *insecure* ciphers are those to which practical exploitation methods have been successfully developed.

This analysis is conducted using the *ssl-enum-ciphers* script, a tool within the Network Mapper (*nmap*) utility (Nmap, 2024). *Nmap* is a widely used open-source tool for network scanning and security assessment, designed to identify hosts and services running on a network. The script systematically tests SSL/TLS connections by attempting to use various ciphers and data compression methods, recording whether the server accepts or rejects each one. The results are then evaluated against the SSL Server Rating Guide (Qualys, 2024b) to assess the server's security configuration.

3.2.2. The trustworthiness of the digital certificate chain

The authenticity of online services can be verified using digital certificates, such as SSL/TLS certificates for web applications. Thus, this assessment dimension aims to examine the certificate chain used by EGOV services to ensure it is correctly configured, valid, and trusted.

Digital certificates comprise electronic documents or container files housing a public key and relevant identifying attributes regarding the entity that controls the corresponding private key (Whitman & Mattord, 2021). Certificates are typically issued by a trusted third party, known as a Certification Authority (CA), and are transmitted to a client during the TLS handshake process with the server provider. The client then uses a Public Key Infrastructure (PKI) to validate the received certificate and verify the service identity. Globally, multiple CAs are organized hierarchically, implying that verifying a certificate may require validating the entire chain up to the top-level entity (i.e., root certificate). A breakdown in this process undermines the trust placed in the certificate's authenticity and consequently compromises the security of an online platform.

The majority of current browsers provide a straightforward way to verify the attributes from a session's certificate and issue alerts if the received certificate presents any problem (i.e., revoked, expired, or signed by an untrusted CA). This functionality also indicates that cryptographic protocols protect the communication between the user's device and the server hosting the accessed service. Nevertheless, it is important to recognize that the perception of security may be deceptive, as browsers do not assess the versions and robustness of the algorithms employed to sign the certificate or encrypt the data being exchanged.

The assessment performed in this stage comprises evaluating multiple properties of digital certificates, namely the strength of the signature algorithm used, the timeframe validity, the correct configuration of host names, and the revocation status. This evaluation relies on the *SSL Server Test* online service (Qualys, 2024c), which establishes SSL/TLS sessions with the analyzed platforms to retrieve all certificates in their trust chain up to the root certificate.

3.2.3. The exposure of the hosting servers to known vulnerabilities

In addition to evaluating the strength of cryptographic protocols (described in Section 3.2.1), all the platforms were analyzed to identify their potential attack surface due to exposure to known exploitable software vulnerabilities.

In many cases, online platforms frequently comprise numerous publicly accessible software components, leading to multiple open ports on public Internet Protocol (IP) addresses. It is not unusual for several online platforms to utilize a shared set of host servers. This increases the potential attack surface (number of accessible components) and, therefore, the number of known and unknown vulnerabilities, leading to a higher probability of achieving successful exploitation of one or more vulnerabilities.

Table 1

Region and sub-region distribution of analysed domains.

Region	Sub-region	2023	2024	Region	Sub-region	2023	2024
Africa	Eastern Africa	208	204	Americas	Caribbean	68	67
	Middle Africa	21	19		Central America	51	49
	Northern Africa	29	27		North America	35	34
	Southern Africa	53	49		South America	115	108
	Western Africa	118	114	Europe	Eastern Europe	192	185
Asia	Central Asia	60	59		Northern Europe	245	240
	Eastern Asia	65	60		Southern Europe	263	258
	South-eastern Asia	391	371		Western Europe	218	216
	Southern Asia	368	352	Oceania	Australia & New Zealand	144	138
	Western Asia	377	351		Melanesia	33	33
					Micronesia	12	10
					Polynesia	2	1

This assessment leverages *Nmap* for probing open ports whilst also identifying services and software version information. Subsequently, the retrieved Common Platform Enumeration (CPE) is used to search for known vulnerabilities associated with them in the public database for Common Vulnerabilities and Exposures (CVEs), managed by the National Institute of Standards and Technology (NIST, 2024a). Additionally, the Common Vulnerability Scoring System (CVSS) is utilized to assess the exposure severity and associated risks. The CVSS consists of an industry-standard framework used to evaluate and communicate the severity of computer systems and software security vulnerabilities. This framework provides a uniform quantitative approach for measuring aspects such as the impact and exploitability of a given vulnerability, ultimately enabling organizations to prioritize and address security issues in an effective manner. In its most recent iteration (version 3), the CVSS attributes vulnerabilities a score of *low*, *medium*, *high*, or *critical* according to their severity.

It is relevant to note that the reconnaissance scans adopted in this work are designed to ensure that information gathering does not access non-public or protected data nor produce a substantial load that could disrupt or interfere with platforms' regular operation. This also means, though, that other identifiable vulnerabilities requiring more intrusive probing were not collected in this study. In most cases, this more intensive probing will not be a concern for a potential attacker. To this point, the data collected for this dimension should be understood as conservative. Moreover, all the assessments are agnostic to the service provided, meaning that this work does not consider the sensitivity of the services hosted on the analyzed domains.

4. Results

This section presents the results of the comparative analysis conducted across the two assessment periods, using the three dimensions outlined in Section 3.2. For clarity, the first and second assessments are

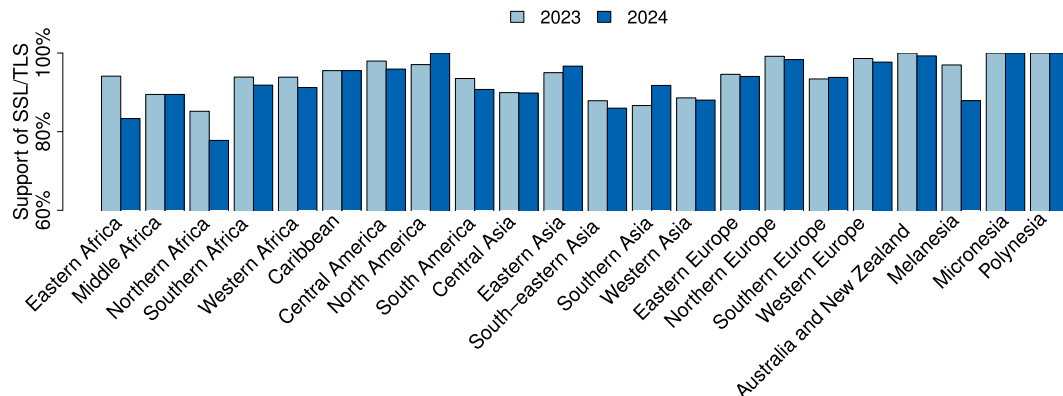
referred to as the 2023 and 2024 assessments, respectively, throughout this section.

4.1. Security in end-to-end communications

Over the period of this study, the number of domains supporting SSL/TLS-based connections decreased from 92.73 % to 91.75 %. This means that 28 domains reachable in both assessments lost the compatibility with any transport layer secure communication. The decline is most significant in Eastern and Northern Africa, as well as Melanesia, which together account for 50 %. While the global trend indicates a decline in support for secure communications, some regions, such as North America, Eastern Asia, and Southern Europe, display a modest yet opposing trend of improvement. The detailed evolution between assessments is depicted in Fig. 2.

Considering the domains supporting secure connections between public services and citizens, the most widely deployed protocol version worldwide remains TLS 1.2 (see Fig. 3). The study observed an uptick from 47.9 % to 57.8 % in the support of TLS 1.3, the most recent and secure specification (Rescorla, 2018). However, these values are still below the 67.8 % global average estimated support by February 2024 (Qualys, 2024a). This scenario might be related to the challenges of upgrading cryptography protocols (Ott et al., 2023). Despite the support for SSL 2.0 and SSL 3.0 being residual, there is still a significant number of domains supporting versions 1.0 and 1.1 of the TLS protocol (25.6 % and 28 %, respectively), even with these versions having been deprecated by the IETF (Housley, 2019) back in 2019. While this study does not extensively explore the underlying causes of the observed trends, the continued support for older TLS versions may be linked to the need for compatibility with legacy systems.

Although over 91 % of the surveyed platforms supported SSL/TLS-based connections by 2024, this does not mean that communications between citizens and public service servers are secure. Maintaining

**Fig. 2.** Evolution of SSL/TLS support by sub-region.

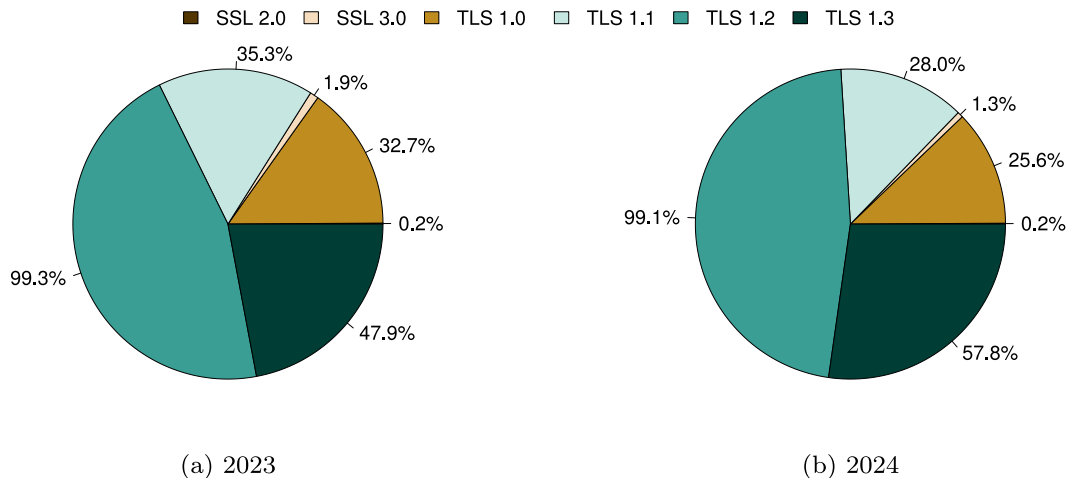


Fig. 3. Global protocol versions' distribution.

support for older and deprecated protocols exposes services and clients to high-risk threats. For instance, according to the National Vulnerabilities Database (NIST, 2024a), the TLS 1.0 protocol is currently exposed to 50 vulnerabilities (16 with CVSS classified as *high* and 2 as *critical*), while the TLS 1.1 protocol is currently exposed to 65 vulnerabilities (33 with a *high* CVSS classification and 3 classified as *critical*).

Most SSL/TLS communications vulnerabilities result from protocols supporting *weak* and *insecure* cryptographic functions. For instance, TLS 1.0 and TLS 1.1 rely on SHA-1 hash, which can be used to perform collision attacks (Leurent & Peyrin, 2020). The MD5 algorithm also represents serious threats of collision and preimage attacks (Wang & Yu, 2005). By supporting these protocols, public services handling sensitive data are highly vulnerable to downgrade attacks, which can enable malicious actors to impersonate the server and execute Man-in-the-Middle (MitM) attacks, intercepting citizens' information. These cryptographic weaknesses undermine the fundamental security guarantees that should protect sensitive communications. Additionally, maintaining support for outdated protocols increases costs, as it requires continuous patching of vulnerabilities and library maintenance and expands the overall attack surface.

By analysing the currently supported protocols, the study revealed that between assessments, the number of platforms using at least one *weak* cipher decreased from 78 % to 71 %, while the platforms still using at least one *insecure* cipher decreased by approximately 1 %. Despite this progress, 110 domains remain exposed due to *insecure* ciphers. Micronesia and Melanesia exhibit the largest reductions in the number of platforms with active *insecure* ciphers, with decreases of 2.2 % and 5.6 %, respectively. The sub-regions with the highest percentages of platforms supporting *insecure* ciphers are Northern Africa (14.3 %), Southern Europe (9.5 %), and South America (7.1 %). These findings suggest that a country's income level does not correlate with the exposure level of cryptographic protocols in use, given that over 99 % of European online platforms are hosted in upper-middle or high-income countries (as presented in Fig. 1).

When considering the number of *weak* and *insecure* ciphers active on each domain, Figs. 4 and 5 show a worrying scenario. Concerning *weak* ciphers, it was observed an overall deterioration mainly related with domains in Asia, Europe, and Oceania. Globally, half of the assessed platforms (more than 1470) had over 10 active *weak* ciphers. Interestingly, though, low-income countries demonstrated an improvement between the two assessment periods (see Fig. 4b).

The global median of *insecure* ciphers per domain, consistently at 2 across both assessments, represents a higher risk despite being lower in number compared to *weak* ciphers. Fig. 5b shows that the distribution of *insecure* ciphers remains relatively uniform across all income levels, with some outliers observed in upper-middle and lower-middle-income

countries.

Even when considering the need for backward compatibility as a justification for extended support of deprecated SSL/TLS protocol versions, the identified security weaknesses reveal a significant gap between current security practices and effective regulatory oversight. Many legislative frameworks explicitly recognize the critical risks posed by weak encryption. The findings highlight that systems administrators should at least phase out *weak* and *insecure* ciphers while reinforcing the importance of evidence-based policymaking, where security assessments of production systems complement regulatory approaches to ensure robust cybersecurity practices.

4.2. Certificate chain trustworthiness

As introduced in Section 3, digital certificates sustain the security model in which citizens can trust the services they are communicating with are authentic. To provide this trust, besides deploying the required SSL/TLS certificates to all services, system administrators must ensure a trusted CA signs them and that all the chain, from the local certificate up to the root certificate, is valid (non-revoked) and uses secure signatures and key sizes.

Although certificates play a fundamental role in providing trust to citizens, Fig. 6 presents another alarming scenario. In 2024, over 72 % of the domains hosting the analyzed platforms presented at least one issue related to their SSL/TLS certificate chains, whilst over 7 % of the issues were in their root certificates. Even considering an improvement regarding 2023, the analysis presented in Fig. 6 shows that there is still considerable room for improvement in this regard.

Analysing the geographic distribution of domains, all regions exhibit improvements in reducing the presence of at least one issue in their certificate chains, with Asia showing the most significant reduction at approximately 10 %. The regions with the greatest improvements are Melanesia (22 %), Northern Africa (17.6 %), and Western Asia (17.6 %). However, this positive trend does not extend to issues in root certificates that saw a slight deterioration in Africa, Europe, and Oceania (see Fig. 6a).

From the countries' income point of view, Fig. 6b depicts a similar trend, with all income levels showing signs of improvement concerning the issues in the local certificate. For the problems in root certificates, the panorama in low-income countries degraded, contrasting with the remaining income levels, where an improvement can be seen. From an economic perspective, the results in Fig. 6b suggest that a country's income level might correlate to the proportion of domains with issues in their certificate chain and root certificate. In the case of low-income countries, even with the improvement from 2023 to 2024, over 94 % of domains presented issues in their chain, a troubling situation given

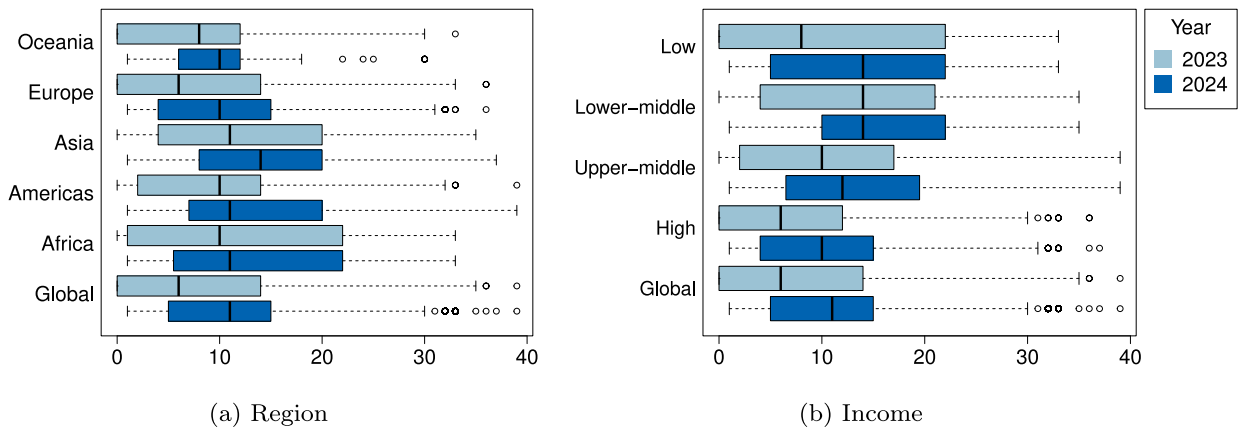


Fig. 4. Distribution weak ciphers per platform.

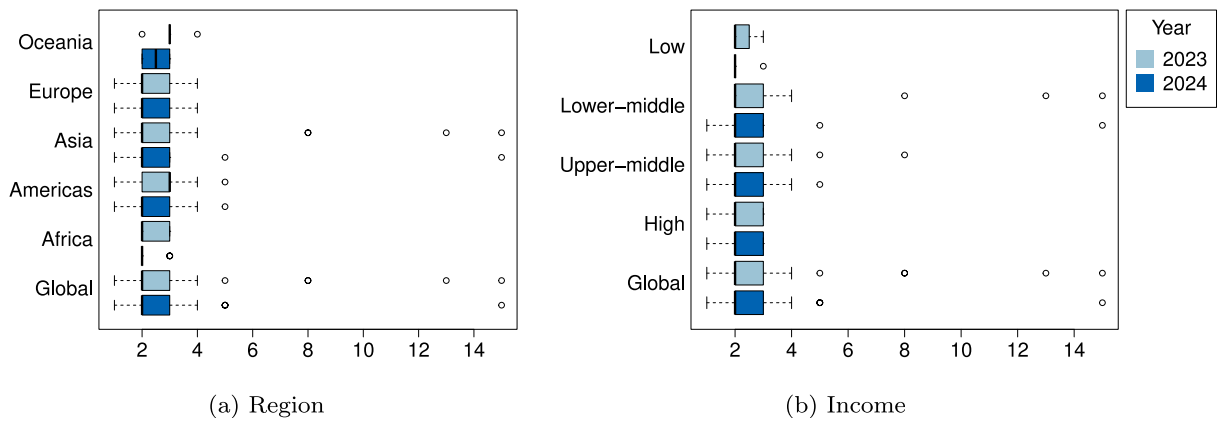


Fig. 5. Distribution of insecure ciphers per platform.

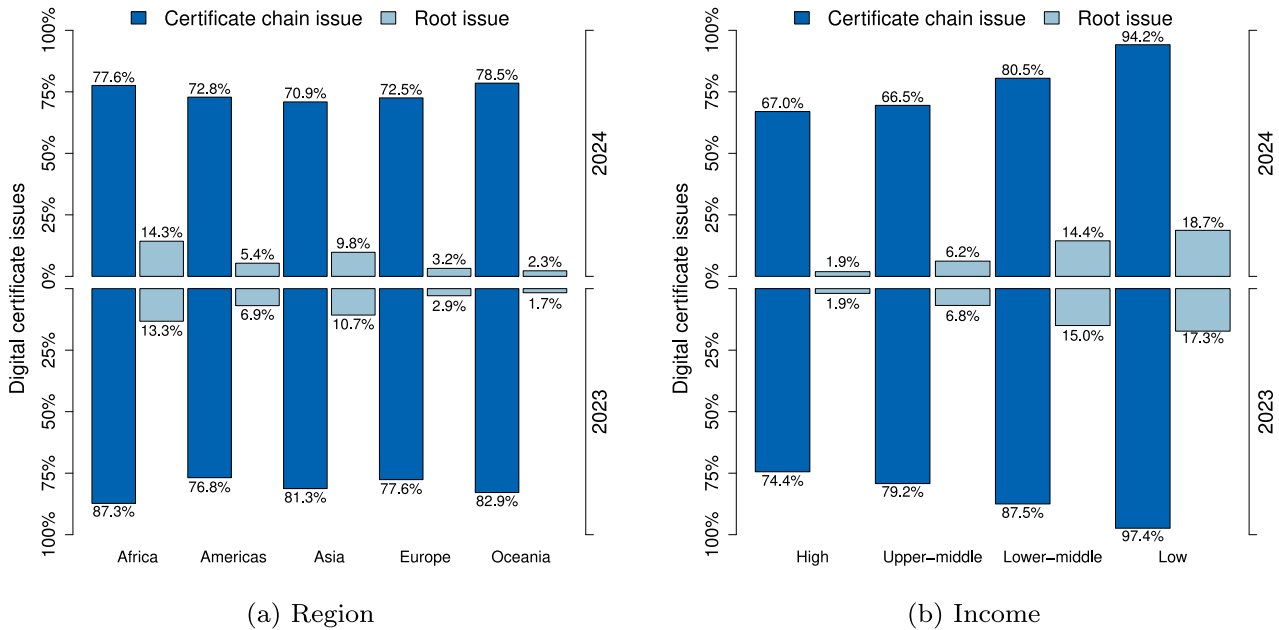


Fig. 6. Evolution of the SSL/TLS certificate chain issues.

that the government representatives who shared the analyzed domains stated them as hosting strategic national EGOV platforms (UN DESA, 2022). This panorama reinforces the importance of international

cooperation and capacity-building initiatives to ensure that economic constraints do not hinder compliance with cybersecurity standards and regulations.

A deeper analysis of these results reveals the most common issues in certificates used by the surveyed platforms. The observed trends are presented by region in Fig. 7 and by income level in Fig. 8.

Globally, the most noticeable change lies in a reduction of about 5 % in the use of *insecure signatures* and an equivalent increase in *hostname mismatch* problems. Yet, *insecure signatures* remain the most common issue globally, meaning either the private key used to sign the certificate or the hashing function used in the signature is *weak* or *insecure*. The analysis also identified the insecure *SHA-1 hashing function* as the primary contributor to these issues. High-income and upper-middle-income countries show the most significant improvements in addressing this problem, reducing the number of domains with *insecure signatures* by 7 % and 5 %, respectively.

A frequent source of certificate-related issues is *hostname mismatches*. This happens when the *Common Name* attribute of a certificate does not match the Uniform Resource Locator (URL) used to reach the service. In these scenarios, attackers can more easily create convincing phishing sites that mimic legitimate government services. Domains of high-income (7 %), upper-middle-income (4 %), and lower-middle-income (4 %) countries showed a deterioration in this dimension when compared to what was observed in 2023. Once again, economic aspects do not seem to be the cause. A possible reason for this decline could be ineffective certificate renewal procedures that fail to consider all the domain names relying on them.

Another recurring issue identified in this study is that many evaluated platforms lacked a valid certificate during both assessments due to expiration. Globally, 873 domains were using an expired certificate by the time of the second assessment (i.e., the certificate attribute *not after* (NAF) was set to a date previous to the assessment). This number increased by 3.3 % between assessments. Regionally, domains in Africa and Oceania showed the highest rates of expired certificates in use, at 38.69 % and 38.26 %, respectively (see Fig. 7). Expired certificates can cause service disruptions as modern browsers block access to websites using them. This leads to operational downtime for critical government services, affecting both citizen access and government operations. Two potential factors could be driving this issue: inadequate management practices for digital certificates and the expenses associated with their issuance.

A notable positive aspect of the digital certificates used by the analyzed public services is the generally low occurrence of *revoked* (RVK) and *blocklisted*¹ (BLK) certificates. Notably, the global incidence of blocklisted certificates dropped to zero, representing an ideal outcome. While there is a slight increase in revoked certificates, their presence remains minimal and confined to Asia at residual levels (see Fig. 7).

Beyond service disruption, improperly configured certificates can lead to loss of trust and reputational damage for government institutions. Hostname mismatches and unverified revocation statuses create opportunities for phishing attacks and identity spoofing, deceiving users into interacting with fraudulent services. Moreover, the absence of automated monitoring and renewal policies increases administrative overhead while leaving vulnerabilities unaddressed for extended periods.

Introducing periodic security audits and public transparency reports can enhance trust and align service providers' practices with international standards and guidelines. More specifically, EGOV service providers should ensure proper certificate chain configuration by using trusted CAs and validating all certificate levels. Enforcing the usage of strong cryptographic protocols is also essential to mitigate security risks related to digital certificates. Automated certificate management and strict renewal policies can prevent expired or misconfigured certificates

while addressing hostname mismatches to ensure proper authentication. Additionally, implementing certificate revocation checks through Certificate Revocation Lists (CRL) and Online Certificate Status Protocol (OCSP) stapling strengthens citizens' trust in the provided services.

4.3. Service provider vulnerability

The final dimension of this study provides an overview of known vulnerabilities in the software running on publicly accessible hosting servers of online platforms. These vulnerabilities often stem from using software components with security flaws or weaknesses, increasing attack surfaces, and exposing both services and citizens to multi-vector threats.

Fig. 9 presents the regional trends in the proportion of domains with at least one known vulnerability. A consistent pattern of decline is evident across all regions, showing a reduction in the number of domains exposed to one or more vulnerabilities. This decrease is particularly pronounced in Middle Africa, Central Asia, and Micronesia. Globally, a positive trend can be observed since, between assessments, the number of exposed domains decreased from just over 20 %, to under 12 %. Nevertheless, this remains a significant issue that needs to be addressed, particularly given the apparent correlation with economic realities. In the 2024 assessment, over 30 % of low-income countries were found to be exposed to at least one vulnerability, compared to less than 5 % of high-income countries.

According to the NIST CVSSv3 classification (NIST, 2024b), which evaluates vulnerability severity based on exploitability and impact, categorized as *low* (CVSS 0.1–3.9), *medium* (CVSS 4.0–6.9), *high* (CVSS 7.0–8.9), and *critical* (CVSS 9.0–10.0), three key trends emerge: (i) a rise in vulnerabilities of *medium* severity, (ii) a decline in those with *high* severity, and (iii) a slight increase in exposure to *critical* vulnerabilities. Fig. 10 provides a comparative view of how exposure levels have changed across different vulnerability severity levels.

There are two contrasting findings in these results. On the one hand, a decrease in *high* severity vulnerabilities can be viewed as a step towards minimizing the exploitation of security flaws or weaknesses. On the other hand, the increase in *critical* vulnerabilities is a worrying scenario due to the higher risk they present to the security of public services and citizens alike. Furthermore, this increase provides a new context to Fig. 9, showing that an overall lower proportion of exposed platforms does not necessarily imply an improvement in the overall security of platforms and citizens.

A more detailed analysis of the CVSS distribution in 2024 is provided in Fig. 11 for both regions and income levels, adding a geo-economic perspective on the exposure severity of online platforms. Oceania, Americas, and Europe are the regions showing a higher concentration of vulnerabilities with *medium* severity, while vulnerabilities with *high* severity are more concentrated in Europe, Africa, and Asia. All regions have a visible presence of *critical* vulnerabilities, a trend that can also be observed among all income levels. The main difference observed between the various income tiers lies in a higher concentration of *medium* severity vulnerabilities in high-income countries, whilst the remaining tiers show a more noticeable presence of *high* severity vulnerabilities. Although the distribution depicted in Fig. 11b suggests a better scenario in high-income countries, the presence of exploitable *critical* vulnerabilities across all income tiers highlights a concerning global risk for online public services.

Table 2 presents the 20 most common vulnerabilities identified in 2023 and 2024 assessments, their severity, and number of affected platforms. Considering the number of platforms exposed, the study shows a visible drop in 2024, an outcome in line with the pattern seen in Fig. 9. Another relevant finding lies in the severity of the most recurring vulnerabilities, specifically *high* and *critical*. In 2023, 12 out of 20 vulnerabilities were classified as *high* or *critical*, whilst in 2024, that number increased to 18. This emphasizes the idea that despite having fewer domains exposed, the identified vulnerabilities possess a higher severity,

¹ This term was adopted in order to comply with ACM's "Words Matter" alternative terminology recommendations, available at <https://www.acm.org/diversity-inclusion/words-matter>.

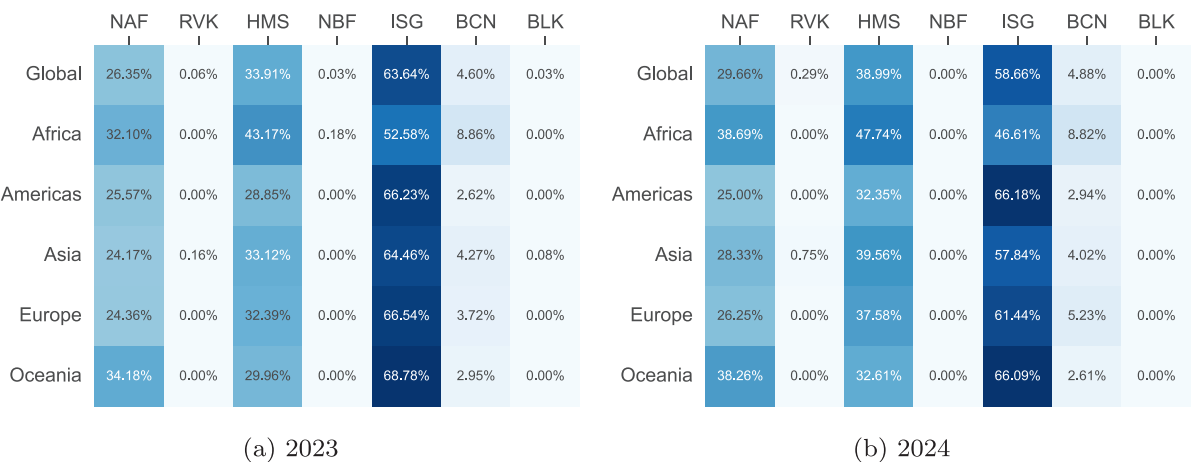


Fig. 7. Evolution of the distribution of certificate problems by region. NAF - Not After, RVK - Revoked, HMS - Hostname Mismatch, NBF - Not Before, ISG - Insecure Signature, BCN - Bad Common Name, BLK - Blocklisted

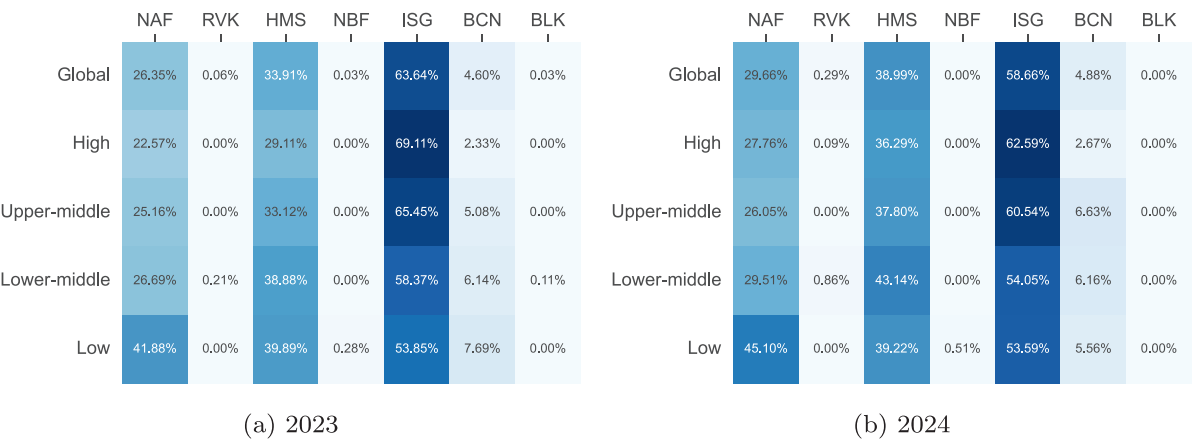


Fig. 8. Evolution of the distribution of certificate problems by income. NAF - Not After, RVK - Revoked, HMS - Hostname Mismatch, NBF - Not Before, ISG - Insecure Signature, BCN - Bad Common Name, BLK - Blocklisted

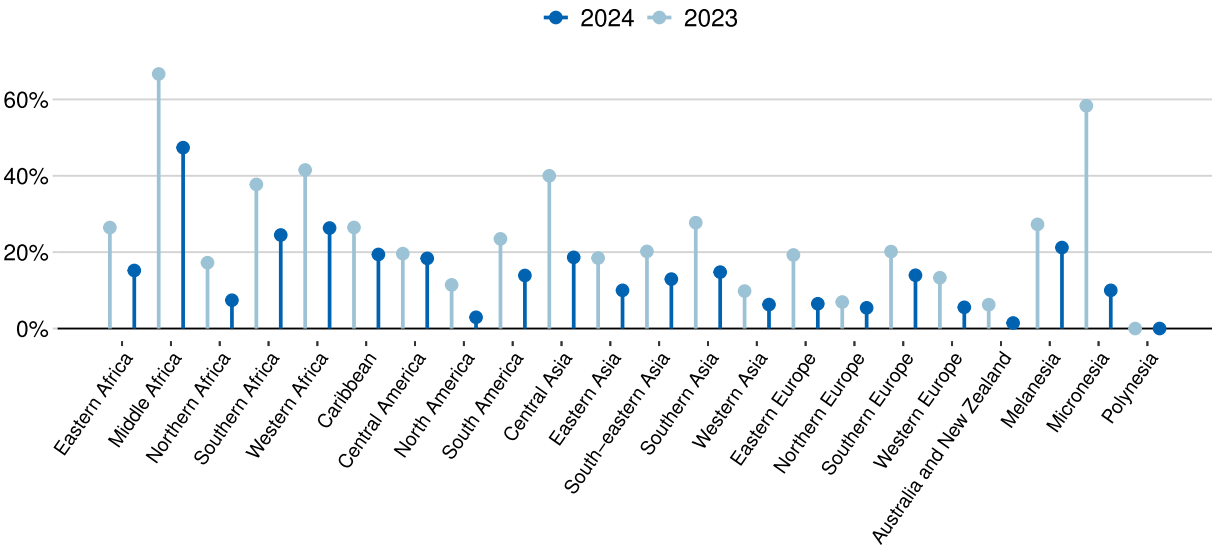


Fig. 9. Global evolution of vulnerability presence across platforms.

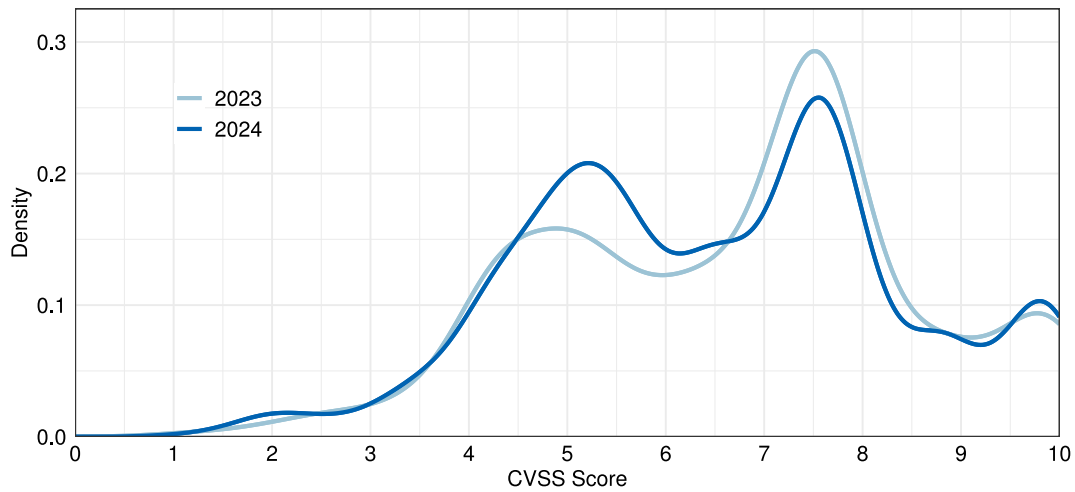


Fig. 10. Global evolution of CVSS distributions.

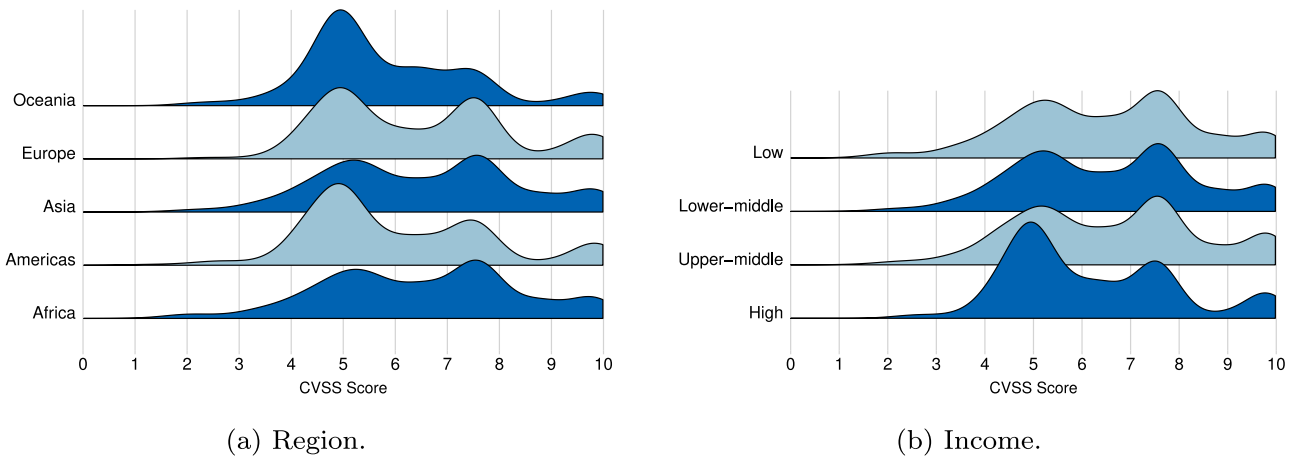


Fig. 11. Distribution of CVSS in 2024.

hence posing a higher security risk. Moreover, around 75 % of the top 20 vulnerabilities in the 2024 assessment were discovered and published before 2023, suggesting that several platforms lack a monitoring and response policy for addressing recent known vulnerabilities. Combined with the previously discussed severity assessment, this further amplifies security risks for both infrastructures and citizens. Recent industry reports indicate that the time between vulnerability discovery and exploitation is rapidly decreasing, with attackers now able to exploit vulnerabilities within an average of 15 days after discovery (Cybersecurity and Infrastructure Security Agency (CISA), 2025).

The decline in known vulnerabilities is an encouraging sign, yet the persistence of critical vulnerabilities suggests that patch management policies remain inadequate. Governments should adopt systematic information security assessments as a proactive tool for policy evaluation and enforcement. Given the number of platforms sharing critical vulnerabilities, these findings reinforce the need to strengthen information-sharing among providers, ultimately enhancing international threat intelligence within the public sector.

5. Discussion

The findings of this study reveal both progress and persistent security gaps in the information security posture of online public services that pose risks to both service providers and citizens. While the adoption of TLS 1.3 has increased over the study period (from 47.9 % to 57.8 %), outdated and insecure protocol versions remain widely supported,

exposing platforms to avoidable risks. Note that using obsolete encryption provides a false sense of security because it may appear that sensitive data is protected when, in reality, it is not. This can lead to complacency among users and system administrators who believe their communications are secure while remaining vulnerable to interception and decryption. This is particularly concerning given the nature of information often shared with EGOV platforms, such as personal identification details, financial information, and other sensitive data.

Similarly, while the proportion of platforms with at least one known vulnerability has decreased from 20 % to under 12 %, the growing presence of *critical* vulnerabilities raises concerns about security practices and long-term risk mitigation strategies. The persistence of known vulnerabilities creates an extended exploitation window for attackers, who can often exploit these weaknesses shortly after they are discovered. This highlights the critical need for timely patch management processes and systematic vulnerability monitoring to reduce exposure risks.

Issues related to digital certificate management remain widespread, with over 72 % of domains exhibiting at least one problem in their certificate chain. This is particularly pronounced in low-income countries, where the correlation between economic constraints and weaker security postures suggests that financial and technical limitations hinder some countries' ability to implement robust security measures. However, this is not a reality restricted to low-income countries. The continued use of insecure cryptographic protocols, expired certificates, and unpatched vulnerabilities call for a more proactive approach to

Table 2
Top 20 vulnerabilities to which platforms are exposed - Global overview.

2023			2024		
CVE	Severity	Platforms exposed	CVE	Severity	Platforms exposed
CVE-2020-14145	Medium	252	CVE-2023-31122	High	166
CVE-2021-41617	High	250	CVE-2022-37436	Medium	139
CVE-2018-15919	Medium	183	CVE-2023-25690	Critical	139
CVE-2022-28614	Medium	181	CVE-2006-20001	High	139
CVE-2022-28615	Critical	181	CVE-2022-36760	Critical	139
CVE-2022-26377	High	181	CVE-2022-28615	Critical	136
CVE-2022-29404	High	181	CVE-2022-26377	High	136
CVE-2022-30556	High	181	CVE-2022-30556	High	136
CVE-2022-31813	Critical	181	CVE-2022-29404	High	136
CVE-2022-23943	Critical	177	CVE-2022-31813	Critical	136
CVE-2022-22719	High	177	CVE-2022-28614	Medium	136
CVE-2022-22720	Critical	177	CVE-2022-22720	Critical	133
CVE-2022-22721	Critical	177	CVE-2022-22719	High	133
CVE-2019-6111	Medium	176	CVE-2022-22721	Critical	133
CVE-2019-6110	Medium	176	CVE-2022-23943	Critical	133
CVE-2018-20685	Medium	176	CVE-2021-44790	Critical	118
CVE-2019-6109	Medium	176	CVE-2021-40438	Critical	117
CVE-2021-44790	Critical	169	CVE-2021-39275	Critical	117
CVE-2019-17567	Medium	166	CVE-2021-34798	High	117
CVE-2021-26690	High	166	CVE-2021-26690	High	116

cybersecurity policies and enforcement worldwide.

These findings extend the work of [Alsmadi and Shanab \(2016\)](#) and [Ali and Zamri Murah \(2018\)](#) by providing a global perspective rather than focusing on single countries like Jordan and Libya. The comprehensiveness of 3068 analyzed platforms across all 193 UN Member States offers a more complete picture of worldwide security practices than the limited sample of 40 websites from Australia and Thailand examined by [Thompson et al. \(2020\)](#).

The observed security gaps align with [Csontos and Heckl \(2021\)](#)

findings in Hungarian public websites, but the present study reveals these issues are pervasive across different regions and income levels. The widespread use of insecure protocols confirms [Silva & Fonte, 2019](#)'s concerns about SSL/TLS adoption in Portugal, demonstrating this is a global rather than localised issue. The results also support [Tegethoff et al. \(2023\)](#) theoretical approaches to secure online services by providing empirical evidence that proper security practices should be integrated into the core design of digital services rather than added as an afterthought. Moreover, the security vulnerabilities identified validate and expand upon the concerns raised by [Shayganmehr et al. \(2023\)](#) and [Li et al. \(2024\)](#) regarding the relationship between data protection and citizen trust. In this sense, the authenticity implications of the widespread problems with digital certificates are critical for services' trustworthiness.

Unlike the interview-based methodologies of [AlGhamdi et al. \(2022\)](#) and [Ali et al. \(2020\)](#), the assessment of production services provide objective diagnostics of security practices, which can reveal discrepancies between stated policies and actual implementations. These empirical findings serve as quantifiable indicators that extend [Pawson's \(2002\)](#) evidence-based policymaking framework, enabling governments to shift from reactive incident reporting towards proactive cybersecurity governance. The systematic assessment methodology demonstrated in this study provides precisely the type of broad, tested, and replicable evidence that Pawson advocated as essential for informing effective policy decisions, moving beyond what [Busetti and Scanni \(2025\)](#) and [de Bruijn and Janssen \(2017\)](#) identified as the limitations of reactive approaches to cybersecurity.

From an operational perspective, EGOV service providers can resort to the described methodology to implement automated certificate management, enforce strict security policies, and integrate real-time vulnerability monitoring to mitigate the level of exposure identified by this work. This must be accompanied by governments integrating regular and systematic assessments into policymaking and enforcement strategies. Evidence-based policies that typically resort to incident reports can be enhanced by the expanded knowledge enabled by understanding the attack surface of production systems providing actual services to citizens. From a strategic dimension, strengthening international cooperation, promoting knowledge-sharing among service providers, and enhancing cybersecurity capacity-building efforts are essential to improving global EGOV security resilience.

6. Conclusions

This study advances the research on EGOV cybersecurity by providing a global analysis of the information security posture of online public services across all 193 UN Member States. While previous research has predominantly focused on isolated countries or regions, this work analyzes 3068 platforms spanning diverse geo-economic contexts, revealing systemic vulnerabilities and misconfigurations that go beyond geographic and economic boundaries. By integrating World Bank economic classifications, the results enable a nuanced analysis of security posture disparities, revealing that economic development alone does not reflect robust security postures, as evidenced by the higher rate of digital certificates insecurely signed in high-income countries. Such findings challenge conventional assumptions about the relationship between national wealth and cybersecurity implementation effectiveness.

Methodologically, this work employs a practical approach combining longitudinal analysis (2023–2024 assessments) with multi-dimensional security evaluations that examine the trustworthiness of digital services being used by citizens and the level of exposition to cyber threats. Unlike theoretical frameworks or questionnaire-based approaches prevalent in existing literature, this empirical methodology provides actionable evidence of actual security practices in production systems. The non-intrusive scanning approach establishes a replicable framework for large-scale security assessments while maintaining

ethical boundaries.

The research relied exclusively on external, observation-based evaluations, deliberately omitting deeper probes requiring authenticated access or internal audits, potentially overlooking organizational-level security measures. It also did not evaluate the specific sensitivity of hosted services, which could provide further insights into risk prioritisation. The findings suggest that authenticated vulnerability assessments would likely reveal a more concerning security landscape. The analysis showed that approximately 75 % of the top 20 vulnerabilities identified in the 2024 assessment were published before 2023, indicating a widespread failure to promptly address known vulnerabilities through adequate monitoring and response policies. This persistence of unfixed vulnerabilities represents systemic weaknesses in patch management and suggests that deeper, authenticated assessments could uncover additional *critical* internal vulnerabilities.

The study also bridges a critical research gap by demonstrating the value of systematic infrastructure assessments as a complement to traditional incident reporting frameworks, providing actionable insights for policymakers and system administrators tasked with securing EGOV platforms. By identifying persistent avoidable vulnerabilities, the findings underscore the importance of integrating regular security evaluations into policy enforcement to ensure compliance with international standards. Targeted investments in cybersecurity capacity-building, automation, and cross-sector collaboration are vital to addressing security gaps, particularly in resource-constrained regions, thereby hardening EGOV infrastructures and minimizing exposure to cyber threats. Furthermore, by addressing these challenges through regular assessments and policy refinement, governments can reinforce the resilience of digital public services, fostering trust and security in EGOV ecosystems globally. The methodology adopted captures the dynamic nature of cybersecurity postures, providing a solid framework for evidence-based policymaking towards enhancing the protection of digital public services.

Future research should explore the underlying organizational and regulatory factors contributing to persistent security gaps in EGOV services. This comprehensive approach might seek to overcome key limitations in current cybersecurity management methodologies by offering standardized metrics for comparisons across different regions and governance frameworks—a crucial advancement given the borderless nature of cyber threats. Studies assessing the impact of policy interventions over time could provide deeper insights into the effectiveness of regulatory frameworks. Additionally, expanding assessments to include authenticated scanning and user experience perspectives could offer a more comprehensive evaluation of the evolving EGOV threat landscape, building upon the new paradigm for global cybersecurity benchmarking established by this work.

CRedit authorship contribution statement

Diogo Ribeiro: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation. **Vítor Fonte:** Writing – review & editing, Writing – original draft, Validation, Funding acquisition. **Luis Felipe Ramos:** Writing – review & editing, Data curation. **João Marco Silva:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is financed by National Funds through the Portuguese

funding agency, FCT - Fundação para a Ciência e a Tecnologia, within project LA/P/0063/2020 and co-financed by Component 5 – Capitalization and Business Innovation, integrated in the Resilience Dimension of the Recovery and Resilience Plan within the scope of the Recovery and Resilience Mechanism (MRR) of the European Union (EU), framed in the Next Generation EU, for the period 2021-2026, within project ATE, with reference 56.

References

- Ainslie, S., Thompson, D., Maynard, S., & Ahmad, A. (2023). Cyber-threat intelligence for security decision-making: A review and research agenda for practice. *Computers & Security*, 132, Article 103352. <https://doi.org/10.1016/j.cose.2023.103352>
- Alarabiat, A., Soares, D., Ferreira, L., & de Sá-Soares, F. (2018). Analyzing e-governance assessment initiatives: An exploratory study. In *Proceedings of the 19th annual international conference on digital government research: Governance in the data age*.
- AlGhamdi, S., Win, K. T., & Vlahu-Gjorgievska, E. (2022). Employees' intentions toward complying with information security controls in Saudi Arabia's public organisations. *Government Information Quarterly*, 39(4), Article 101721. <https://doi.org/10.1016/j.giq.2022.101721>
- Ali, A. A., & Zamri Murah, M. (2018). Security assessment of Libyan government websites. In *2018 cyber resilience conference (CRC)*.
- Ali, O., Shrestha, A., Chatfield, A., & Murray, P. (2020). Assessing information security risks in the cloud: A case study of Australian local government authorities. *Government Information Quarterly*, 37(1), Article 101419. <https://doi.org/10.1016/j.giq.2019.101419>
- Alketbi, A., Nasir, Q., & Talib, M. A. (2018). Blockchain for government services — Use cases, security benefits and challenges. In *2018 15th learning and technology conference (L&T)*.
- Alsmadi, I., & Shanab, E. A. (2016). E-government website security concerns and citizens' adoption. *Electronic Government, an International Journal*, 12, 243–255.
- de Bruijn, H., & Janssen, M. (2017). Building cybersecurity awareness: The need for evidence-based framing strategies. *Government Information Quarterly*, 34(1), 1–7. <https://doi.org/10.1016/j.giq.2017.02.007>
- Busetti, S., & Scanni, F. M. (2025). Evaluating incident reporting in cybersecurity: From threat detection to policy learning. *Government Information Quarterly*, 42(1), Article 102000. <https://doi.org/10.1016/j.giq.2024.102000>
- Chen, S., Hao, M., Ding, F., Jiang, D., Dong, J., Zhang, S., Guo, Q., & Gao, C. (2023). Exploring the global geography of cybercrime and its driving forces. *Humanities and Social Sciences Communications*, 10, 1–10.
- Coleman, S. (2008). Foundations of digital government. In *Digital government: E-government research, case studies, and implementation* (pp. 3–19). Springer.
- Conti, M., Dargahi, T., & Dehghantanha, A. (2018). Cyber threat intelligence: Challenges and opportunities. In *Cyber threat intelligence* (pp. 1–6). Springer.
- Csontos, B., & Heckl, I. (2021). Accessibility, usability, and security evaluation of hungarian government websites. *Universal Access in the Information Society*, 20, 139–156.
- Cybersecurity and Infrastructure Security Agency (CISA). (2025). *Remediate vulnerabilities for internet-accessible systems*. CISA Insights.
- Horowitz, M. C., Allen, G. C., Saravalle, E., Cho, A., Frederick, K., & Scharre, P. (2018). *Artificial intelligence and international security*. Center for a New American Security.
- Housley, R. (2019). *Using Pre-Shared Key (PSK) in the Cryptographic Message Syntax (CMS)*. RFC 8696. URL: [RFC8696](https://www.rfcs.org/rfc/8696).
- Leurent, G., & Peyrin, T. (2020). SHA-1 is a shambles: First chosen-prefix collision on SHA-1 and application to the PGP web of trust. In *29th USENIX conference on security symposium (USENIX security 20)*.
- Li, Y., Yang, R., & Lu, Y. (2024). A privacy risk identification framework of open government data: A mixed-method study in China. *Government Information Quarterly*, 41(1), Article 101916. <https://doi.org/10.1016/j.giq.2024.101916>
- NIST. (2024a). National vulnerability database. URL <https://nvd.nist.gov/>.
- NIST. (2024b). NVD - vulnerability metrics. URL <https://nvd.nist.gov/vuln-metrics/cvss>.
- Nmap. (2024). Network mapper. URL <https://nmap.org/>.
- Ott, D., Paterson, K., & Moreau, D. (2023). Where is the research on cryptographic transition and agility? *Communications of the ACM*, 66, 29–32.
- Pawson, R. (2002). Evidence-based policy: The promise of realist synthesis. *Evaluation*, 8 (3), 340–358. Sage Publications.
- Qualys. (2024a). SSL Pulse. URL <https://www.ssllabs.com/ssl-pulse/>.
- Qualys. (2024b). SSL server rating guide. URL <https://www.ssllabs.com/projects/rating-guide/>.
- Qualys. (2024c). SSL server test. URL <https://www.ssllabs.com/sslttest>.
- Rescorla, E. (2018). *The transport layer security (TLS) protocol version 1.3. RFC 8446*. URL: [RFC8446](https://www.rfcs.org/rfc/8446).
- Shayganmehr, M., Kumar, A., Garza-Reyes, J. A., & Zavadskas, E. K. (2023). A framework for assessing trust in e-government services under uncertain environment. *Information Technology & People*, 36(7), 2718–2755. <https://doi.org/10.1108/ITP-01-2021-0096>
- Silva, J. M. C., & Fonte, V. (2019). Data security and trustworthiness in online public services: An assessment of Portuguese institutions. In *Proceedings of the 12th International Conference on Theory and Practice of Electronic Governance* (pp. 348–353). <https://doi.org/10.1145/3326365.3326411>

- Silva, J. M. C., Ribeiro, D., Ramos, L. F., & Fonte, V. (2024). A worldwide overview on the information security posture of online public services. *Proceedings of the 57th Hawaii International Conference on System Sciences* (pp. 1881–1890).
- Susanto, H., & Almunawar, M. N. (2016). Security and privacy issues in cloud-based E-government. In *Cloud computing technologies for connected government* (pp. 292–321). IGI Global.
- Tegethoff, T., Santa, R., Daú, G., & Cortes, J. (2023). Online security in e-government as an antecedent of cost-effectiveness and quality in business operations. *31(2)*, 125–144. <https://doi.org/10.1108/ICS-10-2021-0159>
- Thompson, N., Mullins, A., & Chongsutakawewong, T. (2020). Does high e-government adoption assure stronger security? Results from a cross-country analysis of Australia and Thailand. *Government Information Quarterly*, 37, Article 101408.
- UN DESA. (2022). *United Nations E-government survey 2022: The future of digital government*. United Nations Department of Economic and Social Affairs.
- UN DESA. (2024). *United Nations E-government survey 2022: Accelerating digital transformation for sustainable development*. United Nations Department of Economic and Social Affairs.
- Wang, X., & Yu, H. (2005). How to break MD5 and other hash functions. In *24th Annual international conference on the theory and applications of cryptographic techniques*.
- Whitman, M. E., & Mattord, H. J. (2021). *Principles of information security*. Cengage learning.

Diogo Pinto Ribeiro is a Ph.D. candidate in informatics at the Department of Informatics in the School of Engineering at the University of Minho, an invited assistant at the same

institution, and an assistant researcher at INESC TEC. His research topics include identity management systems, data consolidation systems, reputation systems, and information security.

Vítor Fonte is an assistant professor in the Department of Informatics, University of Minho (Portugal), and a member of the High Assurance Software Laboratory, a research unit of the same institution and associated with INESC TEC. His research interests are in distributed systems, particularly regarding causality tracking in autonomous operation environments and in distributed key-value stores.

Luis Felipe M. Ramos is a Ph.D. candidate in law at the School of Law at the University of Minho, Portugal. His research is focused on the regulation of emerging technology. His doctoral research investigates the processing of personal data obtained from recently developed biometric technologies, particularly concerning the current legal data protection framework applicable in the EU and the USA.

João Marco Silva is an assistant professor at the Department of Informatics in the School of Engineering at the University of Minho, a Senior researcher at INESC TEC and an adjunct professor at UNU-EGOV. His research topics include information security, network security, and identity management systems. He has been involved in and coordinated multiple research and innovation projects related to non-authoritative identification systems, as well as security modelling and assessment.